WORKED SOLUTION

STEP I 1993, Question 6

Pure • Curve Sketching

> **STEP I 1993, Question 6**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/93-s1-q6

The key idea

Everything turns on how x^N behaves when $N = 10^{100}$. If $|x| < 1$, even a little below 1, then x^N is unimaginably small; if $|x| > 1$, even a little above 1, it is unimaginably large; and at $x = \pm 1$ it is exactly 1 (because N is even). So the fraction $x^N/(1+x^N)$ behaves like a switch: 0 inside $(-1, 1)$ and 1 outside. For h the idea is the opposite one: $10^{-N}x$ is so tiny that sin is indistinguishable from its argument.

Solution

Throughout, write

$$s(x) = \frac{x^N}{1+x^N} = \frac{1}{1+x^{-N}} \quad (x \neq 0), \qquad s(0) = 0,$$

so that $\mathrm{f}(x) = 2 + s(x)$ and $\mathrm{g}(x) = x\,s(x)$.

The features of f

- *Symmetry and range.* N is even, so $x^N = (-x)^N \geq 0$. Hence f is even, and $0 \leq s(x) < 1$ gives $2 \leq \mathrm{f}(x) < 3$.
- *The level $y = 2$ for $|x| < 1$.* Here $x^N \to 0$ with staggering speed: for example $0.99^N = 10^{N \log_{10} 0.99} \approx 10^{-4.4 \times 10^{97}}$. So $s(x) \approx x^N$ is far below anything a graph could show, and $\mathrm{f}(x) = 2$ to any visible accuracy. Exactly, $\mathrm{f}(0) = 2$.
- *The level $y = 3$ for $|x| > 1$.* Now x^{-N} is just as tiny, so $s(x) = 1/(1+x^{-N})$ is 1 to any visible accuracy and $\mathrm{f}(x) = 3$, approached from below. (The line $y = 3$ is in fact an asymptote as $|x| \to \infty$.)
- *The jumps at $x = \pm 1$.* At $x = \pm 1$, $x^N = 1$, so $\mathrm{f}(\pm 1) = \frac{5}{2}$, halfway up. The function is continuous and differentiable, so the vertical segments in the sketch are really an

extremely steep climb. Differentiating,

$$\mathrm{f}'(x) = \frac{Nx^{N-1}}{(1+x^N)^2}, \qquad \mathrm{f}'(1) = \frac{N}{4} = 2.5 \times 10^{99}.$$

Ask yourself How narrow is the "vertical" part of the graph?

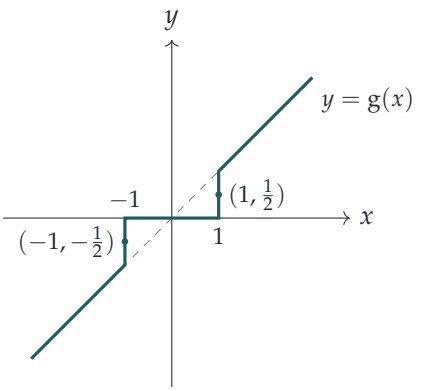
Put $x = 1+\delta$ with δ small, so $x^N = \mathrm{e}^{N\ln(1+\delta)} \approx \mathrm{e}^{N\delta}$ and $s(x) \approx 1/(1+\mathrm{e}^{-N\delta})$. This climbs from 0.01 to 0.99 as $N\delta$ goes from $-\ln 99$ to $\ln 99$, that is, over an interval of width about 9×10^{-100}. On any drawing, that is a vertical line. The same happens at $x = -1$ by symmetry. (The sign of f' also tells us that f increases for $x > 0$ and decreases for $x < 0$, since x^{N-1} has the sign of x.)

The graph of g

Since $\mathrm{g}(x) = x\,s(x)$, the switch s now multiplies the line $y = x$:

- for $|x| < 1$, $s(x)$ is negligible, so $\mathrm{g}(x) = 0$ to any visible accuracy (exactly 0 at $x = 0$);
- for $|x| > 1$, $s(x)$ is 1 to any visible accuracy, so $\mathrm{g}(x) = x$; precisely, $x - \mathrm{g}(x) = \dfrac{x}{1+x^N}$, which is negligible;
- at $x = \pm 1$, $\mathrm{g}(\pm 1) = \pm\frac{1}{2}$, halfway up each jump.

g is odd, because s is even and x is odd. It is also increasing: $\mathrm{g}'(x) = s(x) + x\,s'(x)$, and both terms are ≥ 0 because x and $s'(x)$ have the same sign. At $x = 1$ the gradient is $\mathrm{g}'(1) = \frac{1}{2} + \frac{N}{4}$, so again we see an apparently vertical jump, this time from 0 up to 1 at $x = 1$ and from -1 up to 0 at $x = -1$.



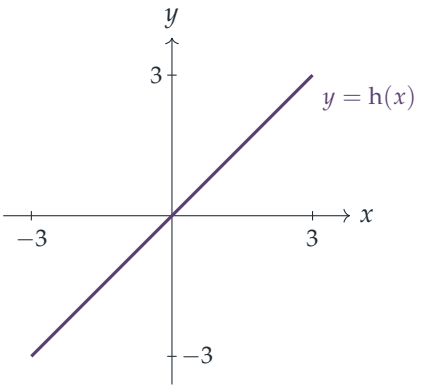
The sketch of g on $[-3, 3]$: zero on $(-1, 1)$, the line $y = x$ (dashed) outside, joined by near-vertical climbs through $(\pm 1, \pm\frac{1}{2})$.

The graph of h

Let $u = 10^{-N}x$. For $|x| \le 3$ we have $|u| \le 3 \times 10^{-N}$, which is extraordinarily small. For small u, $\sin u = u - \frac{u^3}{6} + \cdots$, and for $u > 0$ we have the bounds $u - \frac{u^3}{6} \le \sin u \le u$. Multiplying by 10^N, for $0 \le x \le 3$,

$$0 \le x - \mathrm{h}(x) \le \frac{x^3}{6}\,10^{-2N} \le \frac{27}{6} \times 10^{-2N},$$

and h is odd, so the same holds for $-3 \le x \le 0$ with the signs reversed. So on $[-3,3]$ the graph of h is the straight line $y = x$ through the origin to any conceivable accuracy, lying a whisker below it for $x > 0$ and above it for $x < 0$. Its gradient is $\mathrm{h}'(x) = \cos(10^{-N}x)$, which is 1 to within about 10^{-2N}.



The sine wave is still there, but it is stretched enormously: its first maximum is at $x = \frac{\pi}{2} \times 10^N$, with height 10^N, so the window $[-3,3]$ sees only the tiny straight-looking piece through the origin.

Checks

- Evaluating directly with $s = 1/(1 + \mathrm{e}^{-N\ln|x|})$: $s(0.999)$ and $s(1.001)$ are 0 and 1 to machine precision, and $s(\pm 1) = \frac{1}{2}$ exactly.
- With a modest N such as $N = 10$ or 200, numerical differentiation confirms $s'(1) = N/4$ and $\mathrm{g}'(1) = \frac{1}{2} + N/4$, and a fine grid shows g is increasing on $[-3,3]$. Increasing N only sharpens the steps.
- $10^{100}\sin(10^{-100}x)$ agrees with x to full double precision for random $x \in [-3,3]$, and the scaled-down analogue $M\sin(x/M)$ with $M = 1000$ obeys the bound $0 < x - M\sin(x/M) \le x^3/(6M^2)$.

Common slips

- Forgetting that N is even: x^N is huge and *positive* for $x < -1$, so f is 3 there, not something else.

- Describing the steps as genuine discontinuities. All three functions are continuous (indeed differentiable); the climbs are just too steep to draw.
- Treating h as an oscillating graph. Over $[-3, 3]$ it never gets anywhere near a turning point.

Beyond STEP

The functions $x^n/(1+x^n)$ converge *pointwise*, as $n \to \infty$ through even values, to a step function that is 0 inside $(-1, 1)$, $\frac{1}{2}$ at ± 1 and 1 outside: a sequence of smooth functions whose limit is discontinuous. This is the classic example showing why analysts need *uniform* convergence, a central idea in a first course on real analysis. The function h illustrates the other side: $M\sin(x/M) \to x$ uniformly on bounded intervals as $M \to \infty$. Abbott's *Understanding Analysis* treats both ideas very readably.

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